INTERNSHIP PROJECT REPORT

Title

Traffic Light Controller using Finite State Machine (FSM) in Verilog HDL

Internship Title

VLSI Design and Embedded System

Submitted by

PAWAR VISHNU GANESH

College Name

PES Modern College of Engineering Pune

Internship Organization

InternsElite

Mentor Name:

Mr.Maven

Duration:

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CERTIFICATE

This is to certify that **PAWAR VISHNU GANESH** has successfully completed the internship project titled **"Traffic Light Controller using Finite State Machine (FSM) in Verilog HDL"** under the guidance of **Mr.Maven** at **Internselite** during the internship period from**15 February 2026 to 15 April 2026**.The project work has been carried out satisfactorily and demonstrates a good understanding of digital design concepts using Verilog HDL.

Mentor Signature: _____

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ABSTRACT

This project presents the design and simulation of a Traffic Light Controller using a Finite State Machine (FSM) implemented in Verilog Hardware Description Language (HDL). Traffic light systems are an essential part of modern transportation infrastructure, playing a critical role in ensuring road safety, minimizing congestion, and maintaining smooth traffic flow at intersections.

The proposed system is designed using a Moore Finite State Machine, where the output depends solely on the current state, ensuring stable and predictable operation. The controller consists of four distinct states representing different traffic conditions for North-South (NS) and East-West (EW) directions. Each state corresponds to a specific combination of traffic signals such as Green, Yellow, and Red.

A counter-based timing mechanism is implemented to control the duration of each state. The system transitions sequentially between states based on predefined timing intervals, ensuring proper coordination between the two traffic directions. The design follows Register Transfer Level (RTL) modeling, which enables a clear and structured representation of the system.

A comprehensive testbench is developed to verify the functionality of the design by applying clock and reset signals. The simulation is performed using an online tool, where waveform analysis is used to validate correct state transitions and output behavior. The results confirm that the controller operates accurately and efficiently without any logical errors.

This project provides practical exposure to key digital design concepts such as FSM design, sequential logic, timing control, and Verilog programming. It also highlights the importance of FSM-based controllers in real-world applications. Furthermore, the design can be extended to include advanced features such as traffic density-based control, pedestrian signals, and emergency vehicle handling, making it suitable for future smart traffic systems.

TABLE OF CONTENTS

Sr. No.	Tittle	Page No.
1	Introduction	6
2	Objective	7
3	Methodology	8-9
4	Project Details	10-16
5	Observation	17
6	conclusion	18
7	Bibliography	19

1. Introduction

Traffic light systems are a fundamental component of modern transportation infrastructure, playing a vital role in regulating vehicle movement at road intersections. They help in maintaining an orderly flow of traffic, reducing congestion, and preventing accidents by ensuring that vehicles from different directions move in a controlled and synchronized manner.

With the rapid growth of urbanization and the increasing number of vehicles on roads, efficient traffic management has become a major challenge. Traditional traffic control methods are being replaced by automated digital systems that offer better accuracy, reliability, and flexibility. In this context, digital design techniques using Hardware Description Languages (HDLs) such as Verilog have gained significant importance in implementing real-time control systems.

A Finite State Machine (FSM) is one of the most effective approaches for designing such control systems. FSMs are widely used in digital electronics to model systems that operate in a sequence of predefined states. They provide a structured way to design systems where outputs depend on current states and inputs, making them highly suitable for applications like traffic light control.

In this project, a Traffic Light Controller is designed using a Moore-type FSM, where the outputs depend only on the current state. The system controls traffic signals for two directions: North-South (NS) and East-West (EW). It operates through four distinct states, each representing a specific traffic condition, such as green, yellow, and red signals for each direction.

The design is implemented using Verilog HDL following Register Transfer Level (RTL) modeling. A counter-based timing mechanism is incorporated to control the duration of each traffic signal, ensuring proper synchronization between different states. The system transitions from one state to another based on clock signals, providing a deterministic and predictable behavior.

To verify the correctness of the design, a testbench is developed and simulation is performed using **EDA Playground**. The simulation results are analyzed using waveform outputs, confirming that the traffic light controller operates correctly with proper timing and sequencing.

This project not only demonstrates the practical implementation of FSM concepts but also provides a strong foundation for understanding sequential circuit design, RTL modeling, and digital system verification. It serves as a stepping stone for developing more advanced and intelligent traffic management systems in the future.

2. Objective

The main objective of this project is to design and implement a Traffic Light Controller using a Finite State Machine (FSM) in Verilog HDL. The project aims to develop a reliable and efficient digital system capable of controlling traffic signals at an intersection.

The specific objectives of this project are:

- To understand and apply the concept of Finite State Machines (FSM) in real-world control systems
- To design a traffic light controller using a **Moore FSM architecture**
- To implement the design using Verilog Hardware Description Language (HDL)
- To develop the system using Register Transfer Level (RTL) modeling techniques
- To incorporate timing control using counters for managing signal durations
- To design and implement a testbench for functional verification
- To simulate and analyze the system behavior using **EDA Playground**
- To verify correct state transitions and output responses through waveform analysis
- To gain practical knowledge of digital system design, simulation, and debugging
- To understand how FSM-based controllers are used in real-time applications such as traffic management systems

3. Methodology

The design and implementation of the Traffic Light Controller were carried out in a systematic manner following standard digital design flow and RTL modeling principles. The methodology adopted for this project is described below:

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1. Requirement Analysis
The first step involved understanding the requirements of a traffic light control system for a two-road intersection (North-South and East-West). The sequence of traffic signals and their timing durations were defined. The need for safe and efficient traffic flow was considered while deciding the system behavior.

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2. FSM Design
A Finite State Machine (FSM) was designed to model the operation of the traffic controller. A **Moore FSM** approach was selected, where outputs depend only on the current state.
The system consists of four states:

- S0: NS Green, EW Red
- S1: NS Yellow, EW Red
- S2: NS Red, EW Green
- S3: NS Red, EW Yellow

State transitions were defined in a cyclic manner to ensure continuous operation of the traffic system.

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3. RTL Design
The system was implemented using Register Transfer Level (RTL) modeling in Verilog HDL. The design includes:

- State register to store the current state
- Counter logic to manage timing delays
- Output logic to control traffic signals

The design is synchronous and operates on the rising edge of the clock signal.

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4. Verilog Coding
The traffic light controller was coded using Verilog HDL. Parameters were used to define state values and timing durations. A case statement was used to control state transitions, and combinational logic was used to generate outputs based on the current state.
Proper coding practices were followed to ensure readability, modularity, and correctness of the design.

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5. Testbench Development
A testbench was developed to verify the functionality of the traffic light controller. The testbench generates:

- Clock signal
- Reset signal

Different simulation scenarios were applied to observe system behavior under various conditions.

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6. Simulation
The design was simulated using **EDA Playground**. The simulation allowed observation of state transitions, counter values, and output signals over time.
Waveforms were generated to analyze the behavior of the system.

7. Result Analysis

The simulation results were carefully analyzed to verify:

- Correct state transitions (S0 → S1 → S2 → S3)
- Proper timing delays using the counter
- Accurate output signals for both NS and EW directions

The waveform confirmed that the system operates as expected.

8. Verification

The final design was verified to ensure reliable and error-free operation. All states were checked for correctness, and the system was confirmed to meet the required specifications of a traffic light controller.

4. Project Details

The project focuses on the design and implementation of a Traffic Light Controller using a Finite State Machine (FSM) in Verilog HDL. The system is a sequential digital circuit that controls traffic signals at an intersection by transitioning through predefined states in a cyclic manner.
The controller is designed to manage traffic flow for two directions: North-South (NS) and East-West (EW). The system ensures that only one direction has a green signal at a time, while the other remains in a red state, thereby preventing collisions and ensuring safe traffic movement.
The design is implemented using Register Transfer Level (RTL) modeling, where the system behavior is described in terms of data flow and control signals. The FSM is used to define different traffic conditions and their transitions based on timing constraints.

Overview of Traffic Light Controller
The Traffic Light Controller operates using a **4-state Moore Finite State Machine**, where each state represents a specific traffic condition.
States of Operation

State	NS Signal	EW Signal
S0	Green	Red
S1	Yellow	Red
S2	Red	Green
S3	Red	Yellow

State Transition
The system follows a cyclic sequence of states:
S0 → S1 → S2 → S3 → S0
Each transition occurs after a predefined time delay controlled by a counter.

Inputs and Outputs
Inputs:
- **clk** : Clock signal (controls synchronization)
- **rst** : Reset signal (initializes system)

Outputs:
- **NS[2:0]** : North-South traffic signals (Red, Yellow, Green)
- **EW[2:0]** : East-West traffic signals (Red, Yellow, Green)

Working Principle
The Traffic Light Controller operates based on a clock-driven FSM combined with a timing counter.
- The system starts from an initial state (S0) when reset is activated
- A counter is used to maintain each state for a fixed duration
- After the specified time, the system transitions to the next state
- The output signals change according to the active state

Timing Configuration:
- Green Signal Duration = 10 clock cycles

- Yellow Signal Duration = 3 clock cycles

The Red signal duration is automatically determined by the active state of the opposite direction.

FSM Type

The design is based on a **Moore Machine**, where:

- Output depends only on the current state
- Provides stable and predictable outputs
- Simplifies the design and debugging process

RTL Design Description

The RTL design of the system consists of the following components:

- **State Register:** Stores the current state of the FSM
- **Counter:** Controls the duration of each state
- **Next State Logic:** Determines the next state based on current state and counter
- **Output Logic:** Generates traffic signals based on the current state

The system is synchronous and operates on the rising edge of the clock signal.

Verilog Implementation

The Traffic Light Controller is implemented using Verilog HDL. The design uses:

- Parameterized state definitions
- Case statements for state transitions
- Always blocks for sequential and combinational logic

Verilog Code

```
module traffic_light (
   input clk,
   input rst,
   output reg [2:0] NS,
   output reg [2:0] EW
);

reg [1:0] state;
reg [3:0] count;

parameter S0 = 2'b00,
         S1 = 2'b01,
         S2 = 2'b10,
         S3 = 2'b11;

parameter GREEN_TIME  = 10;
parameter YELLOW_TIME = 3;

always @(posedge clk or posedge rst) begin
   if (rst) begin
      state <= S0;
      count <= 0;
```

```
    end else begin
        count <= count + 1;

        case(state)
            S0: if (count == GREEN_TIME) begin state <= S1; count <= 0; end
            S1: if (count == YELLOW_TIME) begin state <= S2; count <= 0; end
            S2: if (count == GREEN_TIME) begin state <= S3; count <= 0; end
            S3: if (count == YELLOW_TIME) begin state <= S0; count <= 0; end
        endcase
    end
end

always @(*) begin
    case(state)
        S0: begin NS = 3'b001; EW = 3'b100; end
        S1: begin NS = 3'b010; EW = 3'b100; end
        S2: begin NS = 3'b100; EW = 3'b001; end
        S3: begin NS = 3'b100; EW = 3'b010; end
        default: begin NS = 3'b100; EW = 3'b100; end
    endcase
end

endmodule
```

Explanation of Code

The Traffic Light Controller is implemented using Verilog HDL based on a Moore Finite State Machine (FSM). The design consists of a state register, a counter, and output logic to control traffic signals.
The module takes two inputs: clock (clk) and reset (rst). The clock signal synchronizes the operation of the system, while the reset signal initializes the FSM to the default state.
The FSM contains four states represented using a 2-bit register:

- S0: North-South Green, East-West Red
- S1: North-South Yellow, East-West Red
- S2: North-South Red, East-West Green
- S3: North-South Red, East-West Yellow

A counter is used to control the duration of each state. The green signal is maintained for 10 clock cycles, while the yellow signal is maintained for 3 clock cycles. When the counter reaches the defined limit, the FSM transitions to the next state.
The state transitions occur sequentially in the order:
S0 → S1 → S2 → S3 → S0
The output logic is implemented using a combinational always block. Since the design follows a Moore FSM, the outputs depend only on the current state. Based on the active state, appropriate signals are assigned to the North-South (NS) and East-West (EW) traffic lights.
This design ensures proper synchronization, predictable behavior, and safe traffic control by preventing simultaneous green signals in both directions.

Testbench Implementation

A testbench is developed to verify the functionality of the design. It includes:

- Clock generation using toggling signal
- Reset initialization
- Simulation control for observing behavior

Testbench Code:

```
`timescale 1ns/1ps

module tb;

reg clk;
reg rst;
wire [2:0] NS;
wire [2:0] EW;

traffic_light uut (
    .clk(clk),
    .rst(rst),
    .NS(NS),
    .EW(EW)
);

initial begin
    clk = 0;
    forever #5 clk = ~clk;
end

initial begin
    rst = 1;
    #10 rst = 0;
    #300 $finish;
end

initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, tb);
end

endmodule
```

Testbench Code Explanation

The testbench is designed to verify the functionality of the Traffic Light Controller through simulation. It generates the required input signals such as clock and reset. The clock signal is generated using a periodic toggling mechanism, creating a continuous square wave with a fixed time period. This drives the sequential operation of the FSM.

The reset signal is initially activated to initialize the system to its default state. After a short duration, the reset is deactivated, allowing the FSM to begin normal operation.

The testbench runs the simulation for a sufficient amount of time to observe multiple state transitions. This helps in verifying the complete working cycle of the traffic light controller.

Additionally, waveform dump commands are used to record signal activity during simulation. These waveforms are viewed using the simulation tool to analyze the behavior of internal signals such as state, counter, and outputs.

The testbench does not include any hardware logic but acts as a stimulus generator and observation tool, ensuring that the design functions correctly as per the specified requirements.

Simulation Setup

The Traffic Light Controller design was simulated using EDA Playground, an online platform for Verilog HDL simulation. The Icarus Verilog simulator was selected to compile and execute the design and testbench code.

The Verilog design code was written in the design section, while the testbench was used to generate clock and reset signals. The simulation was configured to run for a fixed duration to observe multiple state transitions.

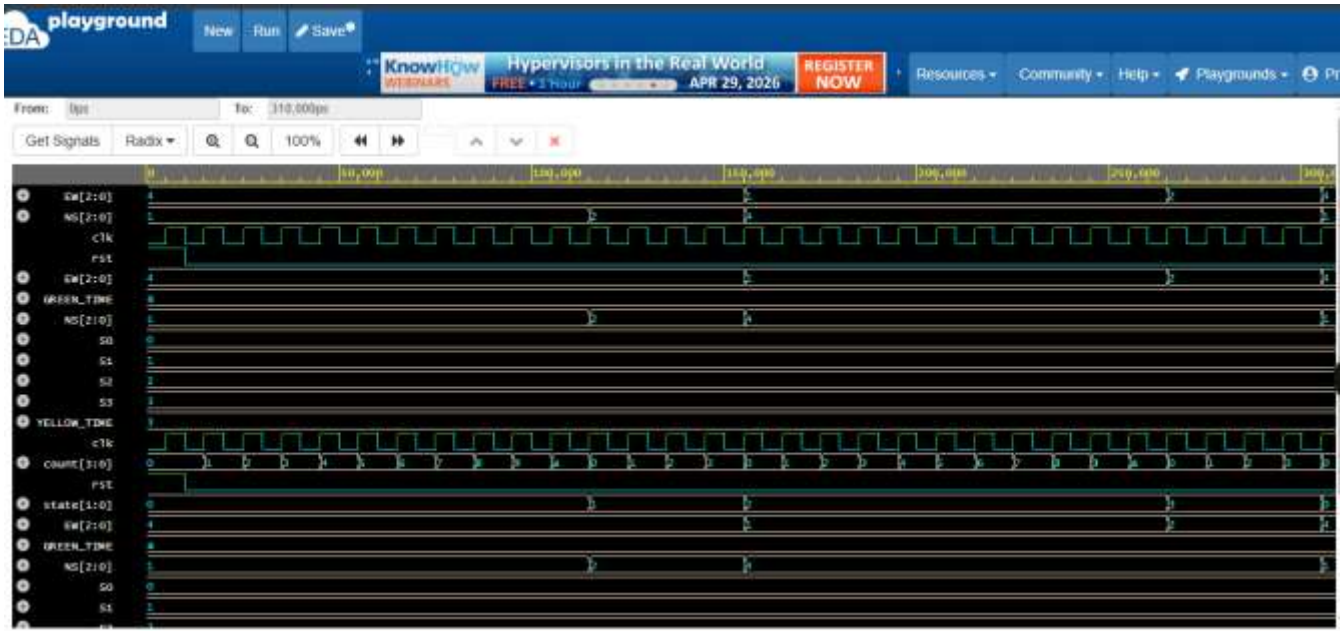
Waveform data was generated using dump file commands and visualized using the EPWave viewer. Key signals such as clock, reset, state, counter, and output signals (NS and EW) were analyzed to verify the correct functionality of the system.

The simulation results confirmed proper state transitions, timing behavior, and correct traffic signal operation.

Waveform Analysis

The simulation waveform demonstrates:

- Sequential state transitions (S0 → S1 → S2 → S3)
- Proper timing using the counter
- Correct output signals for both NS and EW directions
- No overlap of green signals (safe operation)

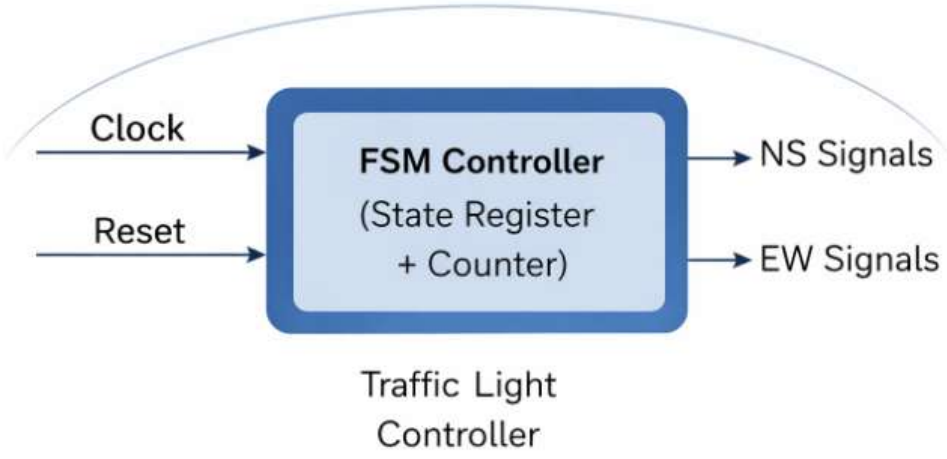


The simulation waveform represents the behavior of the Traffic Light Controller implemented using a Finite State Machine (FSM). The key signals observed in the waveform include clk, rst, state, count, NS, and EW.
At the beginning of the simulation, the reset signal (rst) is activated, which initializes the system to the default state **S0**. In this state, the North-South (NS) direction shows a Green signal (001), while the East-West (EW) direction shows a Red signal (100). This ensures that traffic flows in the NS direction while EW traffic is stopped.
As the clock signal (clk) toggles, the counter (count) starts incrementing. When the counter reaches the predefined value for the green signal duration (10 clock cycles), the FSM transitions from **S0 to S1**. In state **S1**, the NS signal changes to Yellow (010), indicating that the signal is about to change, while EW remains Red.
After the yellow duration (3 clock cycles), the FSM transitions to **S2**, where the NS signal becomes Red (100) and the EW signal turns Green (001). This allows traffic flow in the East-West direction while stopping the North-South traffic.
Similarly, after the green duration for EW, the FSM moves to **S3**, where the EW signal changes to Yellow (010) and NS remains Red. This acts as a warning before switching traffic direction again.
Finally, after the yellow duration, the FSM transitions back to **S0**, completing one full cycle. This sequence continues repeatedly, ensuring continuous and controlled traffic flow.
The waveform clearly shows:

- Correct state transitions (**S0 → S1 → S2 → S3 → S0**)
- Proper timing control using the counter
- Correct output signals for both NS and EW directions
- No overlap of green signals, ensuring safe operation

Thus, the simulation verifies that the Traffic Light Controller operates correctly according to the designed FSM logic.

Block Diagram



5. Observation

The simulation results obtained from the waveform analysis confirm the correct operation of the Traffic Light Controller designed using a Finite State Machine (FSM).

The following observations were made:

- The system initializes correctly to the default state (S0) when the reset signal is activated
- The FSM transitions sequentially through all four states:
  S0 → S1 → S2 → S3 → S0
- The counter successfully controls the duration of each state, maintaining proper timing for green and yellow signals
- The North-South (NS) and East-West (EW) traffic signals change accurately according to the current state
- At any given time, only one direction has a green signal, ensuring safe traffic operation
- The yellow signal appears for a short duration before transitioning between green and red, indicating proper warning behavior
- No glitches or unexpected transitions were observed in the waveform
- The system operates synchronously with the clock signal, ensuring stable and predictable performance
- The design successfully follows Moore FSM characteristics, where outputs depend only on the current state

The waveform clearly demonstrates that the Traffic Light Controller functions correctly with proper sequencing, timing, and output behavior.

6. Conclusion

The Traffic Light Controller using a Finite State Machine (FSM) was successfully designed and implemented using Verilog HDL. The project demonstrates the practical application of digital design concepts such as sequential logic, state machines, and timing control.

A Moore FSM architecture was used to ensure stable and predictable outputs, where the traffic signals depend only on the current state. The system was designed with four states to efficiently manage traffic flow for both North-South and East-West directions.

The use of a counter for timing control enabled accurate duration for green and yellow signals, ensuring proper synchronization between state transitions. The simulation results verified that the system operates correctly, with smooth transitions and no conflicting signals.

The waveform analysis confirms that the controller follows the desired sequence and maintains safe traffic conditions by preventing simultaneous green signals in both directions.

Overall, this project provided a clear understanding of FSM design, RTL modeling, and digital system simulation. It also highlights how such designs can be applied in real-world traffic management systems.
In future, this system can be enhanced by incorporating features such as traffic sensors, pedestrian crossing signals, and adaptive timing mechanisms to make it more efficient and intelligent.

7. Bibliography

- The following resources were referred to during the design and implementation of the Traffic Light Controller project:
- GeeksforGeeks – Verilog HDL and FSM concepts
- Xilinx Documentation – Digital design and simulation references
- EDA Playground – Online simulation platform for Verilog code
- Digital Design textbooks and lecture notes provided by the college
- Online tutorials and reference materials on Finite State Machine (FSM) design